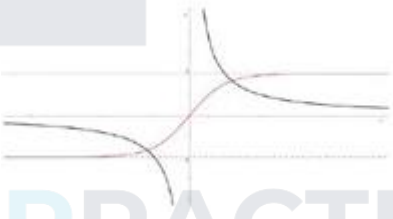


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses quotient or product rule to obtain correct derivative	AO1.1b	B1	$y = \frac{x}{\cosh x} \Rightarrow \frac{dy}{dx} = \frac{\cosh x - x \sinh x}{\cosh^2 x}$
	Clearly sets 'their' $\frac{dy}{dx}$ numerator equal to 0	AO2.4	R1	Stationary point $\Rightarrow \frac{dy}{dx} = 0$ $\Rightarrow \frac{\cosh x - x \sinh x}{\cosh^2 x} = 0$
	Rearranges to complete a rigorous argument to show the required result. AG	AO2.1	R1	$\Rightarrow \cosh x - x \sinh x = 0$ $\Rightarrow \frac{\sinh x}{\cosh x} = \frac{1}{x}$ $\Rightarrow \tanh x = \frac{1}{x}$ AG
(b)	Sketches $\tanh x$ correctly including asymptotes	AO1.2	B1	
	Sketches $\frac{1}{x}$ correctly	AO1.2	B1	
	Deduces correct number of stationary points FT 'their' sketch in (b)(i)	AO2.2a	B1F	
(c)	Finds the second derivative	AO1.1a	M1	$\frac{d^2 y}{dx^2}$
	Obtains a correct expression for the second derivative	AO1.1b	A1	$= \frac{\cosh^2 x (\sinh x - x \cosh x - \sinh x)}{\cosh^4 x}$ $- \frac{2 \cosh x \sinh x (\cosh x - x \sinh x)}{\cosh^4 x}$
	Deduces that the second term is zero by using results from part (a)	AO2.2a	R1	second term is zero at stationary points

			$\frac{d^2y}{dx^2} = -\frac{x}{\cosh x} = -y$ $\Rightarrow \frac{d^2y}{dx^2} + y = 0$ AG
Completes a rigorous argument to show the required result. AG Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	
Total 10 marks			

Q2.

(a) $x = \tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$

$$xe^y + xe^{-y} = e^y - e^{-y}$$

$$\text{or } xe^{2y} + x = e^{2y} - 1$$

M1

$$\Rightarrow (x+1)e^{-y} = e^y(1-x)$$

$$\Rightarrow (x+1) = e^{2y}(1-x)$$

A1

$$e^{2y} = \frac{1+x}{1-x} \Rightarrow y = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

AG

A1cso

(b) $y = \frac{1}{2} \ln(1+x) - \frac{1}{2} \ln(1-x)$

M1

$$\frac{dy}{dx} = \frac{1}{2(1+x)} + \frac{1}{2(1-x)}$$

A1

$$= \frac{1-x+1+x}{2(1+x)(1-x)} = \frac{2}{2(1-x^2)} = \frac{1}{1-x^2}$$

AG

A1cso

Alternative 1



$$\frac{dy}{dx} = \frac{1}{2} \times \frac{(1-x)}{(1+x)} \times \frac{d}{dx} \left(\frac{1+x}{1-x} \right) \quad M1$$

$$\frac{dy}{dx} = \frac{1}{2} \times \frac{(1-x)}{(1+x)} \times \frac{(1-x) + (1+x)}{(1-x)^2} \quad A1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1-x^2} \quad A1 \text{ cso}$$

3

$$(c) \int 4x \tanh^{-1} x \, dx = 4x \tanh^{-1} x - \int \frac{4x}{1-x^2} \, dx$$

M1

$$4x \tanh^{-1} x + 2 \ln(1-x^2)$$

A1

$$\tanh^{-1} \frac{1}{2} = \frac{1}{2} \ln 3$$

must simplify logarithm to $\ln 3$

B1

$$\text{Value of integral} = \ln 3 + 2 \ln \frac{3}{4}$$

any correct form

A1

$$\ln \left(\frac{3^3}{2^4} \right)$$

all working must be correct

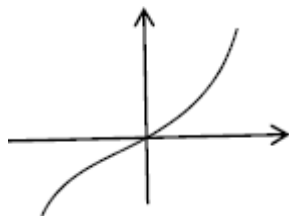
A1cso

5

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[11]

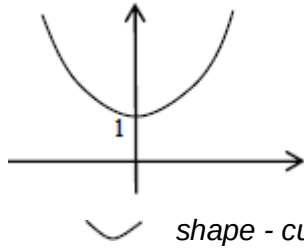
Q3.

(a) (i) $\sinh x$ graph

shape – curve through
O,
in 1st and 3rd quadrants

M1

cosh x graph



M1

Gradient of $\sinh x > 0$ at origin **and** cosh x minimum at (0,1)

A1

3

(ii)

cosh $x = 0$ has **no** solutions

and $\sinh x = -k$ has **one** solution

(hence equation has exactly one solution)

or cosh $x > 0$ etc

(since $y = -k$ cuts $y = \sinh x$ exactly once)

E1

1

(b) $\frac{dy}{dx} = 6\cosh x + 2\cosh x \sinh x$
one term correct

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M1

all correct – may have $6\cosh x + \sinh 2x$

A1

(2) $\cosh x(3 + \sinh x) = 0$

therefore C has only one stationary point

{ putting = 0, factorising
and concluding statement (may be later)

E1✓

$\Rightarrow \sinh x = -3$

finding $\sinh x$ from “their” equation

m1

$$\cosh^2 x = 10$$

$$y (= -18 + 10) = -8$$

*answer must be integer so do not accept
calculator approximation rounded to -8*

A1

5

[9]

Q4.

(a) (i) $\frac{d}{du} (2u\sqrt{1+4u^2}) = \frac{8u^2}{\sqrt{1+4u^2}} + 2\sqrt{1+4u^2}$

*M1 for clear use of product rule
(condone one error in one term)*

M1

correct unsimplified

A1

$$\frac{d}{du} (\sinh^{-1} 2u) = \frac{2}{\sqrt{1+4u^2}}$$

B1

$$\frac{8u^2 + 2}{\sqrt{1+4u^2}} = \frac{2(1+4u^2)}{\sqrt{1+4u^2}} = 2\sqrt{1+4u^2}$$

be convinced – must see this line OE

$$\frac{d}{du} (2u\sqrt{1+4u^2} + 4\sinh^{-1} 2u) = 4\sqrt{1+4u^2}$$

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all working must be correct (not enough to just say k = 4)

A1cso

4

(ii) $\frac{1}{\text{“their” } k} [2u\sqrt{1+4u^2} + \sinh^{-1} 2u]_0^1$

anti differentiation

M1

$$= \frac{\sqrt{5}}{2} + \frac{1}{4} \sinh^{-1} 2$$

FT “their” k or even use of k

A1✓

2

(b) (i) $y = \frac{1}{2} \cos 4x$ and $\frac{dy}{dx} = A \sin 4x$
 $\frac{dy}{dx} = -2 \sin 4x$

substituted into $\int K y \left(1 + \left(\frac{dy}{dx} \right)^2 \right) (dx)$
clear attempt to use formula for CSA

M1

$(S =) \int_0^{\frac{\pi}{8}} 2\pi \times \frac{1}{2} \cos 4x \sqrt{1 + 4 \sin^2 4x} \, dx$ = printed answer

(combining $2 \times \frac{1}{2}$)

AG $\frac{dy}{dx} = -2 \sin 4x$ **and** $2 \times \frac{1}{2}$ **and** dx must
 be seen to award A1cso

A1cso

2

(ii) $u = \sin 4x \Rightarrow du = 4 \cos 4x \, dx$
condone $du = B \cos 4x \, dx$ for M1

M1

$(S =) \frac{\pi}{4} \int_0^1 \sqrt{1 + 4u^2} \, (du)$

condone limits seen later

A1

use of their result from (a)(ii) **correctly** FT "their" B

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 m1

$(S =) \frac{\pi\sqrt{5}}{8} + \frac{\pi}{16} \sinh^{-1} 2$
 OE

A1cso

4

[12]

Q5.

(a) $\frac{dy}{dx} = \frac{1}{2 \tanh x}$

B1

$\times \operatorname{sech}^2 x$

B1

$$= \frac{1}{2 \sinh x \cosh x}$$

for expressing in terms of $\sinh x$ and $\cosh x$

M1

$$= \frac{1}{\sinh 2x}$$

AG; PI by previous line

A1

4

$$(b) \quad \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{1}{\sinh^2 2x}}$$

use of formula; accept $\sqrt{\quad}$ inserted at any stage

M1

$$= \sqrt{\frac{\cosh^2 2x}{\sinh^2 2x}}$$

relevant use of $\cosh^2 - \sinh^2 = 1$

M1

$$= \frac{\cosh 2x}{\sinh 2x}$$

OE

A1

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Integral is $\frac{1}{2} \ln \sinh 2x$

M1 for $\ln \sinh$

M1A1

$$\sinh(2 \ln 4) = \frac{255}{32} \quad \sinh(2 \ln 2) = \frac{15}{8}$$

PI

B1B1

$$s = \frac{1}{2} \ln \left(\frac{17}{4} \right)$$

ft error in $\frac{1}{2}$

A1F

8

Q6.

- (a) Use of $\cosh 2x = 2\cosh^2 x - 1$
or $\cosh 4x = 2\cosh^2 2x - 1$

M1

$$\text{RHS} = \frac{1}{2} \cosh 2x + \frac{1}{2} \cosh^2 2x$$

A1

$$= \frac{1}{4} (1 + 2\cosh 2x + \cosh 4x)$$

A1

3

If substituted for both $\cosh 4x$ and $\cosh 2x$ in LHS M1 only, until corrected

If RHS is put in terms of e^x

M1 for correct substitution

A1 for correct expansion

A1 for correct result

- (b) $\frac{dy}{dx} = 2\cosh x \sinh x = \sinh 2x$

$$\text{allow A1 for } 1 + \left(\frac{dy}{dx}\right)^2 = 1 - 4\cosh^2 x + 4\cosh^4 x$$

Incorrect form for $\cosh^2 x$ in terms of $\cosh 2x$ M1 only

M1A1

Or

$$y = \left(\frac{e^x + e^{-x}}{2} \right)^2 = \frac{e^{2x} + 2 + e^{-2x}}{4}$$

$$\frac{dy}{dx} = \frac{2e^{2x} - 2e^{-2x}}{4}$$

(M1)

$$= \sinh 2x$$

(A1)

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \sinh^2 2x = \cosh^2 2x$$

AG

A1

3

(c) $S = 2\pi \int_{(0)}^{(\ln 2)} \cosh^2 x \cosh 2x dx$
allow even if limits missing

M1A1

$$= 2\pi \int_0^{\ln 2} \frac{1}{4} (1 + 2\cosh 2x + \cosh 4x) dx$$

m1

$$= \frac{2\pi}{4} \left[x + \frac{2\sinh 2x}{2} + \frac{\sinh 4x}{4} \right]$$

Integrated correctly

A1

Correct use of limits $a = 128$, $b = 495$

*accept correct answers written down with no working.
 Only one A1 if 2π not used*

m1
A1,A1

7

[13]

Q7.

(a) $\frac{dy}{dx} = 12 \sinh x - 8 \cosh x - 1$

*The B1 and M1 could be in reverse order
 if put in terms of e first*

B1

$$12 \frac{(e^x - e^{-x})}{2} - 8 \frac{(e^x + e^{-x})}{2} - 1 = 0$$

M0 if $\sinh^x$ and $\cosh^x$ in terms of e^x are interchanged

M1

$$2e^{2x} - e^x - 10 = 0$$

ft slips of sign

A1F

$$(2e^x - 5)(e^x + 2) = 0$$

ft provided quadratic factorises

M1A1F

$$e^x \neq -2$$

some indication of rejection needed

E1

$$x = \ln \frac{5}{2} \text{ one stationary point}$$

A1F

7

$e^x = \frac{5}{2}$
 Condone $e^x = \frac{5}{2}$ with statement provided
 quadratic factorises

Special Case

$$\frac{dy}{dx} = 12\sinh x - 8\cosh x$$

B0

If $\frac{dy}{dx}$ For substitution in terms of e^x M1

leading to $e^{2x} = 5$ A1

Then M0

(b)
$$b = 12 \frac{\left(\frac{5}{2} + \frac{2}{5}\right)}{2} - 8 \frac{\left(\frac{5}{2} - \frac{2}{5}\right)}{2} - \ln \frac{5}{2}$$

 for substitution into original equation

M1A1F

$$= \frac{174}{10} - \frac{84}{10} - \ln \frac{5}{2}$$

CAO

A1

$$= 9 - a$$

AG; accept $b = 9 - a$

A1

4

[11]

Q8.

(a) (i) Divide $\cosh^2 t - \sinh^2 t = 1$ by $\cosh^2 t$

$$\text{Or } \frac{\sinh^2 t}{\cosh^2 t} + \frac{1}{\cosh^2 t}$$

M1

Rearrange

AG If solved back to front with no
 conclusion ending $\cosh^2 t - \sinh^2 t = 1$

B1 only

A1

$$(ii) \quad \frac{d}{dt} \left(\frac{\sinh t}{\cosh t} \right) = \frac{\cosh^2 t - \sinh^2 t}{\cosh^2 t}$$

M1A1

$$= \operatorname{sech}^2 t$$

AG

A1

3

$$(iii) \quad \frac{d}{dt} (\operatorname{sech} t) = -(\cosh t)^{-2} \sinh t$$

Allow A1 if negative sign missing

M1A1

$$= -\operatorname{sech} t \tanh t$$

AG

A1

3

$$(b) \quad (i) \quad \left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 = \operatorname{sech}^4 t + \operatorname{sech}^2 t \tanh^2 t$$

Allow slips of sign before squaring for this M1

M1

Use of $\tanh^2 t + \operatorname{sech}^2 t = 1$

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m1

$$= \operatorname{sech}^2 t$$

A1

$$\therefore s = \int_0^{\frac{1}{2} \ln 3} \operatorname{sech} t \, dt$$

AG (including limits)

A1

4

$$(ii) \quad u = e^t \quad du = e^t \, dt$$

B1

$$\int \operatorname{sech} t \, dt = \int \frac{2}{u^2 + 1} \, du$$

CAO M1 for putting integrand
in terms of u (no $\operatorname{sech}(\ln u)$)

M1A1

$$[2 \tan^{-1} u]$$

Or $2 \tan^{-1} e^t$

A1

Change limits correctly or change back to t
At some stage

m1

$$= \frac{2\pi}{3} - \frac{2\pi}{4} = \frac{\pi}{6}$$

CAO

A1

6

[18]

Q9.

(a) $\frac{dz}{dx} = 2 \cosh x \sinh x$

Any correct method

M1

$= \sinh 2x$
AG

A1

2

(b) $I = \int_{x=0}^{x=1} \frac{dz}{1+z^2}$

Ignore limits here

M1A1

$$= [\tan^{-1} z]_{x=0}^{x=1}$$

A1

$$= [\tan^{-1}(\cosh^2 x)]_0^1$$

Or A1 for change of limits

A1

$$= \tan^{-1}(\cosh^2 1) - \tan^{-1}(\cosh^2 0)$$

$$= \tan^{-1}(\cosh^2 1) - \frac{\pi}{4}$$

AG

A1

5

[7]

Q10.

(a)
$$\frac{d}{dx} \left(\cosh^{-1} \frac{1}{x} \right) = \frac{1}{\sqrt{\frac{1}{x^2} - 1}} \left(-\frac{1}{x^2} \right)$$

$\frac{dy}{dx} = f(y)$
 M0 if $\frac{dy}{dx} = f(y)$ and no attempt to substitute back to x

M1A1

$$= \frac{-1}{x\sqrt{1-x^2}}$$

AG

A1

3

(b) (i)
$$\frac{d}{dx} \left(\sqrt{1-x^2} \right) = \frac{-2x}{2\sqrt{1-x^2}}$$

For numerator

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B1

For denominator (not $(1-x^2)^{\frac{1}{2}}$)

B1

$$\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^2}} + \frac{1}{x\sqrt{1-x^2}}$$

$$= \frac{1-x^2}{x\sqrt{1-x^2}} = \frac{\sqrt{1-x^2}}{x}$$

For attempt to put over a common denominator

M1

AG

A1

4

$$(ii) \quad s = \int_{\frac{1}{4}}^{\frac{3}{4}} \sqrt{1 + \frac{1-x^2}{x^2}} \, dx = \int_{\frac{1}{4}}^{\frac{3}{4}} \frac{1}{x} \, dx$$

For use of $\sqrt{f + \left(\frac{dy}{dx}\right)^2}$

M1A1A1

$$= \left[\ln x \right]_{\frac{1}{4}}^{\frac{3}{4}}$$

M1

$$= \ln \frac{3}{4} - \ln \frac{1}{4} = \ln 3$$

AG

A1

5

[12]

Q11.

$$(a) \quad \frac{dy}{dx} = \frac{1}{\tanh \frac{x}{2}} \dots$$

B1

$$\operatorname{sech}^2 \frac{x}{2} \dots$$

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B1

$$\frac{1}{2}$$

B1

$$= \frac{1}{2 \frac{\sinh \frac{x}{2}}{\cosh \frac{x}{2}} \cosh^2 \frac{x}{2}}$$

OE ie expressing in $\sinh \frac{x}{2}$ and $\cosh \frac{x}{2}$

M1

$$= \frac{1}{2 \sinh \frac{x}{2} \cosh \frac{x}{2}}$$

$$= \frac{1}{\sinh x}$$

ie use of $\sinh 2A = 2 \sinh A \cosh A$

M1

$$= \operatorname{cosech} x$$

AG

A1

Alternative

$$\ln \sinh \frac{x}{2} - \ln \cosh \frac{x}{2}$$

(B1)

$$\frac{1}{2} \frac{\cosh \frac{x}{2}}{\sinh \frac{x}{2}} - \frac{1}{2} \frac{\sinh \frac{x}{2}}{\cosh \frac{x}{2}}$$

(B1B1)

$$\frac{\cosh^2 \frac{x}{2} - \sinh^2 \frac{x}{2}}{2 \sinh \frac{x}{2} \cosh \frac{x}{2}}$$

(M1)

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Use of $\sinh 2A = 2 \sinh A \cosh A$

(M1)

result

(A1)

6

(b) (i) $s = \int_1^2 \sqrt{1 + \operatorname{cosech}^2 x} \, dx$

M1

$$= \int_1^2 \coth x \, dx$$

AG

A1

2

(ii) $s = [\ln \sinh x]^2$
needs to be correct

M1

$$= \ln \sinh 2 - \ln \sinh 1$$

A1

$$= \ln \frac{2 \sinh 1 \cosh 1}{\sinh 1}$$

must be seen

A1F

$$= \ln (2 \cosh 1)$$

AG

A1

4

[12]

Q12.

(a) (i) $\frac{d}{dt} \left(\frac{1}{\cosh t} \right) = -1(\cosh t)^{-2} \sinh t$

$$\frac{-2(e^t - e^{-t})}{(e^t + e^{-t})^2}$$

 Or

M1A1

$$= -\operatorname{sech} t \tanh t$$

AG

A1

3

(ii) Use of $\tanh^2 t = 1 - \operatorname{sech}^2 t$

M1

Printed result

A1

2

(b) (i) $\dot{x} = 1 - \operatorname{sech}^2 t$ ($\dot{y} = -\operatorname{sech} t \tanh t$)

B1

$$\dot{x}^2 + \dot{y}^2 = (1 - \operatorname{sech}^2 t)^2 + \operatorname{sech}^2 t - \operatorname{sech}^4 t$$

Any form

M1A1

$$= 1 - \operatorname{sech}^2 t = \tanh^2 t$$

AG

A1

4

(ii) $s = \int_0^t \tanh t \, dt$

Ignore limits for M1 and first A1

M1

$$= [\ln \cosh t]_0^t$$

A1

$$= \ln \cosh t$$

AG

A1

3

(iii) $e^s \cosh t$

M1

$$y = e^{-s}$$

AG

A1

2

(c) $S = 2\pi \int_0^t \operatorname{sech} t \tanh t \, dt$

Ignore limits for M1 and first A1

M1

$$= 2\pi [-\operatorname{sech} t]_0^t$$

A1

$$= 2\pi (1 - \operatorname{sech} t)$$

A1

$$= 2\pi (1 - e^{-s})$$

AG

A1

4

Q13.

$$(a) \quad (i) \quad 2 \left(\frac{e^{\theta} - e^{-\theta}}{2} \right) \left(\frac{e^{\theta} + e^{-\theta}}{2} \right) = \frac{e^{2\theta} - e^{-2\theta}}{2} = \sinh 2\theta$$

AG

M1A1

2

$$(ii) \quad \left(\frac{e^{\theta} - e^{-\theta}}{2} \right)^2 + \left(\frac{e^{\theta} + e^{-\theta}}{2} \right)^2$$

M1

$$= \frac{e^{2\theta} - 2 + e^{-2\theta} + e^{2\theta} + 2 + e^{-2\theta}}{4}$$

A1

$$= \cosh 2\theta$$

AG

A1

3

$$(b) \quad (i) \quad \dot{x} = 3 \cosh^2 \theta \sinh \theta$$

Allow M1 for **reasonable** attempt at differentiation,
but M0 for putting in terms of $e^{k\theta}$ or $\sinh 3\theta$

unless real progress made towards $\dot{x}^2 + \dot{y}^2$

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M1

$$\dot{y} = 3 \sinh^2 \theta \cosh \theta$$

Allow this M1 if not squared out, must be clear

sum in question is $\dot{x}^2 + \dot{y}^2$

A1

$$\dot{x}^2 + \dot{y}^2 = 9 \cosh^4 \theta \sinh^2 \theta + 9 \sinh^4 \theta \cosh^2 \theta$$

AG

M1

$$= 9 \sinh^2 \theta \cosh^2 \theta (\cosh^2 \theta + \sinh^2 \theta)$$

$$\text{Accept } \int_0^1 \sqrt{\frac{9}{4} \sinh^2 2\theta \cosh 2\theta} d\theta$$

but limits must appear somewhere

A1

$$= \frac{9}{4} \sinh^2 2\theta \cosh 2\theta$$

AG

A1

6

(ii) $S = \int_0^1 \frac{3}{2} \sinh 2\theta \sqrt{\cosh 2\theta} d\theta$

M1

$$u = \cosh 2\theta \quad du = 2 \sinh 2\theta d\theta$$

M1A1

$$I = \int \frac{3}{4} u^{\frac{1}{2}} du = \frac{3}{4} \times \frac{2}{\frac{3}{2}} u^{\frac{3}{2}}$$

A1F

$$S = \left\{ \frac{1}{2} (\cosh 2\theta)^{\frac{3}{2}} \right\}_0^1$$

A1F

$$= \frac{1}{2} \left\{ (\cosh 2)^{\frac{3}{2}} - 1 \right\}$$

AG

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A1

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6

[17]

Q14.

(a) (i) $\frac{e^k + e^{-k}}{2} - \frac{3(e^k - e^{-k})}{2} = -1$

*Allow if 2's are missing or if coshx and
sinh x interchanged*

M1

$$-2e^k + 4e^{-k} = -2$$

A1

$$e^{2k} - e^k - 2 = 0$$

AG Condone x instead of k

A1

3

(ii) $(e^k + 1)(e^k - 2) = 0$

M1

$$e^k \neq -1$$

*Must state something to earn E1.
Do not accept ignoring or crossing out.*

E1

$$e^k = 2$$

A1

$$k = \ln 2$$

A1F

4

(b) (i) $\cosh x = 3\sinh x$ or in terms of e^x

M1

$$\tanh x = \frac{1}{3} \text{ or } 2e^x = 4e^{-x}$$

A1

$$x = \frac{1}{2} \ln \left(\frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} \right) \text{ or } e^{2x} = 2$$

A1F

$$x = \frac{1}{2} \ln 2$$

CAO

A1

4

(ii) $\frac{dy}{dx} = \sinh x - 3\cosh x$ or $-e^x - 2e^{-x}$

M1

$$= 0 \text{ when } \tanh x = 3 \text{ or } e^{2x} = -2$$

A1

Correct reason

Must give a reason

E1

3

$$\frac{d^2y}{dx^2} = y = 0 \text{ at } \left(\frac{1}{2}\ln 2, 0\right)$$

(iii) ie one point

B1F

1

[15]



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Examiner reports

Q2.

- (a) Most students used $x = \tanh y$ and the given exponential form to obtain an expression in terms of x for e^{2y} and hence the printed result. Quite a few students thought that $\tanh^{-1}y = \frac{\sinh^{-1}y}{\cosh^{-1}y}$ and made no progress. Those students who started with $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ were also rarely successful.
- (b) The majority of students used the chain rule to differentiate the result from part (a). Those with greater insight used the law of logarithms to obtain the difference of two logarithms and the new form was then much easier to differentiate. Poor algebraic manipulation meant that some were forced to fudge their working to obtain the printed answer.
- (c) This part of the question was the best source of marks for most students, though some were unable to integrate $\frac{x}{1-x^2}$ correctly. Those whose calculus was sound had no trouble expressing the final answer in the given form.

Q3.

- (a) (i) The graphs of $\sinh x$ and $\cosh x$ usually had the correct basic shape; however, some drew the $\sinh x$ curve with vertical asymptotes or with a zero gradient at the origin and several students failed to indicate that the minimum point of $y = \cosh x$ was at $(0,1)$.
- (ii) Very few students gave a satisfactory explanation here with many referring to the two graphs drawn in part (a)(i) having a “single point of intersection”, which is not true.
- It was necessary to state that the equation $\cosh x = 0$ has no solutions and that, by considering the graph of $\sinh x$, the equation $\sinh x = -k$ has exactly one solution. It was good to see students referring to $\sinh x$ as being a one-one-function, but many failed to write a suitable conclusion with regard to the given equation having a single solution.
- (b) Those who differentiated the given expression for y usually obtained the correct derivative. Some tried to use the double angle formula for $\cosh 2x$ but, if they did this incorrectly, then they were not eligible for full marks even if they obtained the correct expression $6\cosh x + \sinh 2x$. Some used the result from part (a)(ii) to deduce that C has only one stationary point. Others reinvented the wheel by stating that $\cosh x \neq 0$ and then showed that $\sinh x = -3$, before concluding that the curve had a single stationary point. The vast majority failed to make any statement of this form and lost a mark. Those who tried to convert into exponential form usually made no progress beyond the first two marks for differentiation. Having found that $\sinh x = -3$, it was expected that students would have shown that $\cosh^2 x = 10$ by using the identity $\cosh^2 x = 1 + \sinh^2 x$, but many were very dependent on their calculator and some did not provide enough working to convince the examiners that the y -coordinate was an integer. At least one student solved the problem effectively by writing the

equation of the curve as $y = (\sinh x + 3)^2 - 8$.

Q4.

- (a) (i) The differentiation caused more difficulties than had been anticipated. Many students did not use the product rule on the first expression and those who did rarely had both terms correct. Even the derivative of $\sinh^{-1} 2u$ was done badly with many omitting the “2” in the numerator.
- (ii) Those who did not obtain $k = 4$ were still able to score full marks here if they used their value of k or even “ k ” in their anti-differentiation.
- (b) (i) Students needed to show the correct value of $\frac{dy}{dx}$ before substituting into the formula for the curved surface area in order to score full marks. Despite the printed answer, quite a few omitted the limits or the dx or failed to show the product $2 \times \frac{1}{2}$ in their working and lost an easy accuracy mark.
- (ii) Once again the integration by substitution caused problems for many who omitted the factor $\frac{\pi}{4}$ or who did not have the correct limits for u . Because of errors in finding the value of k in part (a)(i) only the best students scored full marks here. Some credit was given for using their previous results correctly even though students did not have the correct value of k .

Q5.

Overall, responses to this question were good and there were many completely correct or nearly correct solutions. Part (a) was usually correct, although lack of intermediate working made it difficult to decide whether a candidate really saw what to do, or merely wrote down the printed answer.

In part (b) the commonest errors were to either think that $\sqrt{1 + 1/\sin^2 2x}$ was equal to $1 + 1/\sin 2x$, or to lose the $2x$ in the process of simplification and end up with $\coth x$ instead of $\coth 2x$. Many candidates arriving at $\int \coth 2x dx$ were able to perform the integration correctly, although a few lost the factor $\frac{1}{2}$.

Q6.

There were many approaches to the first part of this question, but few completely correct solutions. Some students expressed both sides of the identity in terms of exponential functions in order to show them equal and these were largely successful. Many others worked with either the RHS or the LHS of the identity, often ending up with $\frac{1}{2} \cosh^2 2x + \frac{1}{2} \cosh 2x$ but were unable to complete it convincingly. Part (b) was generally answered well, although not all recognised the identity $2 \sinh x \cosh x = \sinh 2x$ which in turn led to some lengthy solutions. Part (c) was also answered well, with many completely correct solutions, and errors, when they did occur, were largely errors of sign or arithmetic slips.

Q7.

There were many good responses to this question. Failure, when it arose, was in the use of methods which did not use the exponential forms for $\sinh x$ and $\cosh x$. Such methods

usually

involved the squaring of the expression for $\frac{dy}{dx}$, more often than not incorrectly. When the squaring was done correctly, there was almost invariably no check that the values obtained were

in fact solutions for $\frac{dy}{dx} = 0$, an essential step as squaring had taken place. Part (b) was often correctly worked and with adequate reasoning.

Q8.

Part (a) was a source of good marks for almost all candidates. If errors did occur they were usually errors of sign. Part (b) was also generally well done although it was disappointing to see the square root of $\text{sech}^2 t \tanh^2 t + \text{sech}^4 t$ written as $\text{sech } t \tanh t + \text{sech}^2 t$ a significant number of times. Responses to part (b)(ii) were mixed. Poor algebraic manipulation in the handling of $\text{sech } t$ when expressed in terms of u let many candidates down badly so that they ended up with a polynomial in u to integrate.

Q9.

Part (a) was usually correctly done. However, responses to part (b) were mixed. It was

clear that not all candidates were familiar with $\int \frac{du}{1+u^2}$, even though it is given on page 8 of the formulae booklet. A significant number of candidates gave this integral as $\ln(1 + \cosh^4 x)$.

Q10.

Few candidates were able to supply a correct proof of part (a). The derivative of $\cosh^{-1}$

$\left(\frac{1}{x}\right)$ was almost invariably given as $\frac{1}{\sqrt{\left(\frac{1}{x}\right)^2 - 1}}$ (followed by the correct answer after a small amount of spurious algebra. Candidates fared better in part (b)(i), although it was a little surprising how many candidates were unable to differentiate $\sqrt{1-x^2}$ correctly. Those candidates who continued and attempted part (b)(ii) frequently produced a correct solution. Solutions to this part which went awry were those in which candidates wrote

$\sqrt{1 + \left(\frac{\sqrt{1-x^2}}{x}\right)^2}$ followed by $\sqrt{1 + \frac{1-x}{x}}$ (which incidentally gave the correct answer!), or

in some cases where candidates arrived at $\sqrt{\frac{1}{x^2}}$ or $\sqrt{x^{-2}}$ but were unable to take the square root correctly.

Q11.

Although many candidates were able to write down $\frac{1}{\tanh \frac{x}{2}}$ multiplied by $\frac{1}{2} \text{sech}^2 \frac{x}{2}$,

fewer were able to combine these results to obtain $\operatorname{cosech} x$. Even those candidates who expressed $\frac{dy}{dx}$ entirely in terms of $\cosh \frac{x}{2}$ and $\sinh \frac{x}{2}$ seemed to balk at the algebra

$$\frac{1}{2 \sinh \frac{x}{2} \cosh \frac{x}{2}}$$

which led to

Part (b)(i) was done well and many candidates were able to arrive at $s = \ln \sinh 2 - \ln \sinh 1$ in part (b)(ii) but were unable to reach the printed answer. If the integral of $\coth x$ was performed incorrectly, it was often by $\coth x$ being replaced by $\frac{1}{\tanh x}$ followed by $\ln \tanh x$ or $\ln \cosh x$ as the integral.

Q12.

Generally, candidates scored quite well on this question. Some candidates struggled with part (a)(i) by not realising that $\operatorname{sech} t$ was $(\cosh t)^{-1}$, instead expressing $\operatorname{sech} t$ in exponential form.

This latter method rarely led to a correct solution. However, apart from some sign fudging in part (a)(ii), most candidates were able to recover to answer parts (a)(ii) and (b)(i) correctly. Very few candidates were able to score the three available marks in part (b)(ii), by either ignoring the limits of integration completely or by writing $s = \ln \cosh t + c$ with no effort to show that the value of c was zero. In spite of some inelegant methods, part (b)(iii) was usually answered correctly.

Part(c) caused problems, often by candidates attempting to integrate $e^{-s} \tanh t$ with respect to t by regarding e^{-s} either as a constant or else as e^{-t} . Of those who correctly integrated to arrive at $-\operatorname{sech} t$, again very few candidates recognised the need for limits and so were unable to arrive at the printed result.

Q13.

Candidates were generally well drilled in proving the identities of parts (a)(i) and (a)(ii) although in (a)(ii) sometimes $(e^{\theta} - e^{-\theta})^2$ was written as $e^{2\theta} + e^{-2\theta}$ with the same result from $(e^{\theta} + e^{-\theta})^2$, thus obtaining the correct answer from incorrect algebra. Part (b)(i) was usually quite well done so long as candidates did not write $\cosh^3 \theta$ as $\frac{1}{8} (e^{\theta} + e^{-\theta})^3$.

Those who worked in powers of e^{θ} and $e^{-\theta}$ found it impossible to reconcile their formula with the printed result and so make little meaningful progress. Part (b)(ii) proved to be

beyond all but the most able candidates. The required integral, $\int \frac{3}{2} \sinh 2\theta \sqrt{\cosh 2\theta} d\theta$, was tackled successfully by these candidates by a variety of methods. Some spotted the integral, others used the substitution $u = \cosh 2\theta$ whilst others again integrated by parts.

Q14.

Again, this question proved to be a good source of marks for many candidates. Some candidates in part (a)(i) mixed the exponential forms for $\cosh x$ and $\sinh x$, whilst others, having expressed $\cosh x$ and $\sinh x$ in exponential form correctly and having arrived at $-e^k$

$+2e^{-k} = -1$ were unable to take the final step which led to the printed result. There was, also, not always a very convincing reason for the rejection of $e^k = -1$ in this part of the question. A very common error in part (a)(ii) was to write $2k - k - \ln 2 = 0$ after the printed answer, leading to the correct answer by totally incorrect mathematics. Part (b)(i) was usually answered well, as was part (b)(ii), apart from those candidates who thought that the derivatives of hyperbolic functions followed the pattern of the derivatives of trigonometrical functions and so incurred sign errors. The explanation for the rejection of solutions to $e^{2x} = -2$ was more convincing in this part of the question than was the rejection of $e^k = -1$ in part (a)(ii). Part (b)(ii) was usually correct.



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